

ACME CORPORATION

Payroll Interface Specification

Document ID: INT-PAY-2027 | Version: 1.0 | Effective: January 1, 2027

1. Overview

This specification defines the data exchange between the Acme Benefits platform and the ADP payroll system for benefit deduction management.

2. Deduction Code Mapping

Benefit	Deduction Code	Tax Treatment	Frequency
Medical PPO 500	MED-PPO-500	Pre-tax (Section 125)	Per paycheck
Medical HDHP 3000	MED-HDHP-3000	Pre-tax (Section 125)	Per paycheck
Dental Basic	DEN-BASIC	Pre-tax (Section 125)	Per paycheck
Dental Premium	DEN-PREM	Pre-tax (Section 125)	Per paycheck
Vision Standard	VIS-STD	Pre-tax (Section 125)	Per paycheck
Voluntary Life	LIFE-VOL	Post-tax	Per paycheck
STD Employee Share	DIS-STD-EE	Post-tax	Per paycheck
LTD Employee Share	DIS-LTD-EE	Post-tax	Per paycheck
HSA Contribution	HSA-CONT	Pre-tax	Per paycheck
FSA Contribution	FSA-CONT	Pre-tax	Per paycheck

3. Pay Period Calculation

All monthly benefit costs are converted to per-paycheck amounts using the following divisors:

Pay Frequency	Annual Pay Periods	Monthly Divisor	Example (\$150/month)
Weekly	52	4.333	\$34.62
Biweekly	26	2.167	\$69.23
Semi-Monthly	24	2.000	\$75.00
Monthly	12	1.000	\$150.00

Acme Corporation uses Biweekly payroll. All benefit deductions are calculated using the Biweekly divisor of 2.167.

4. Deduction Reconciliation

A three-way reconciliation is performed monthly between:

- Benefits system: configured employee contribution amounts
- Payroll system: actual deduction amounts per employee per paycheck
- Carrier: billed premium amounts

Reconciliation matching key: Employee ID (primary) + Plan Code (secondary)

Tolerance: Differences of \$0.02 or less per paycheck are acceptable rounding errors.

Differences greater than \$0.02 must be flagged for review.

(AI DEMO - PAYROLL MISMATCH SCENARIO)

Employee: John Smith (EE-001)

Plan: Medical PPO 500 (MED-PPO-500), Employee Only tier

Expected deduction: \$75.00 per paycheck (based on \$150/month employee share)

Actual deduction: \$200.00 per paycheck

Variance: \$125.00 per paycheck -- EXCEEDS TOLERANCE

AI should identify: Payroll system may still be using prior year rate or wrong coverage tier.

5. Required Payroll File Fields

Field	Type	Description
EmployeeID	String	EE-NNN format
DeductionCode	String	Plan deduction code
ExpectedAmount	Decimal	Benefits system configured amount
ActualAmount	Decimal	Payroll system deduction amount
PayFrequency	Enum	Biweekly / Semi-Monthly / Monthly
EffectiveDate	Date	YYYY-MM-DD
CoverageMonth	String	YYYY-MM
ReconciliationStatus	Enum	Matched / Mismatch / Pending

6. Error Scenarios

Error Code	Description	Action Required
PAY-MISMATCH	Deduction amount does not match configur	HR review within 2 days
PAY-MISSING	Employee enrolled but no deduction found	Payroll team to add
PAY-EXTRA	Deduction found but employee is not enro	Payroll team to remove
PAY-TERM	Deduction continues after termination da	Immediate correction